



launch-assist-sim

RESEARCH SIMULATOR · GROUND-POWERED LAUNCH ASSIST

How much rocket propellant can a ground-powered push replace?

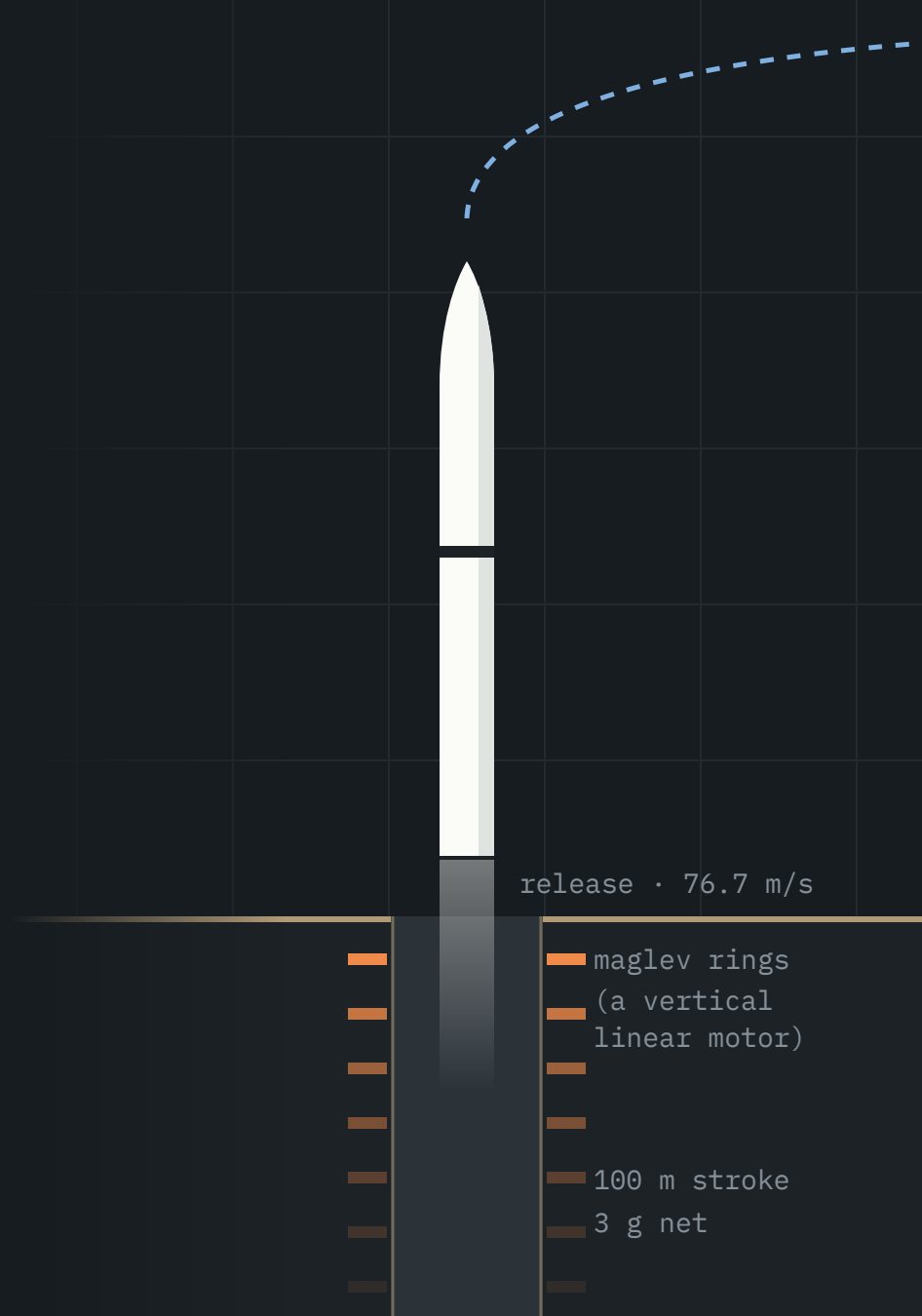
A vertical maglev silo pushes a Falcon 9-class rocket out of the ground at about 77 m/s, before or while its engines light. The simulator measures what that is worth, always against the same rocket launched from a pad.

PRELIMINARY RESULTS

PHASES 0-2 AND SP1 DONE · SP2 NEXT

STATUS AS OF 4 OCTOBER 2026

Rahul Singh Khokhar · October 2026



THE QUESTION

Keep the payload and the orbit. How much propellant can the ground replace?

At a fixed payload and the same 200 km circular orbit, what share of a Falcon 9-class rocket's propellant can an electrically powered silo push replace?

WAY 1 · MORE PAYLOAD

Same propellant, more payload

Measured in Phase 2 on the 2-D model: **+1,498.8 kg** for the cold-start silo (preliminary, slide 10).

WAY 2 · LESS PROPELLANT

Same payload, less propellant

The headline question.

Measured in SP1: **41.26 t** of stage-1 propellant, 10.04% (preliminary, slides 13–15).

WHY A LAUNCH COMPANY MIGHT CARE

- A payload gain helps only on missions that need more mass to orbit. Propellant replaced at a fixed payload would apply at every launch.
- The ground energy can come from renewables and be reused every launch (the hypothesis, README).
- "Energy is cheap; power isn't": the push needs about 1.2 MWh, delivered in 2.6 s at a GW-class peak (README hand calculation).

This is motivation, not a result. Whether any operator wants the trade has not been asked or tested here.

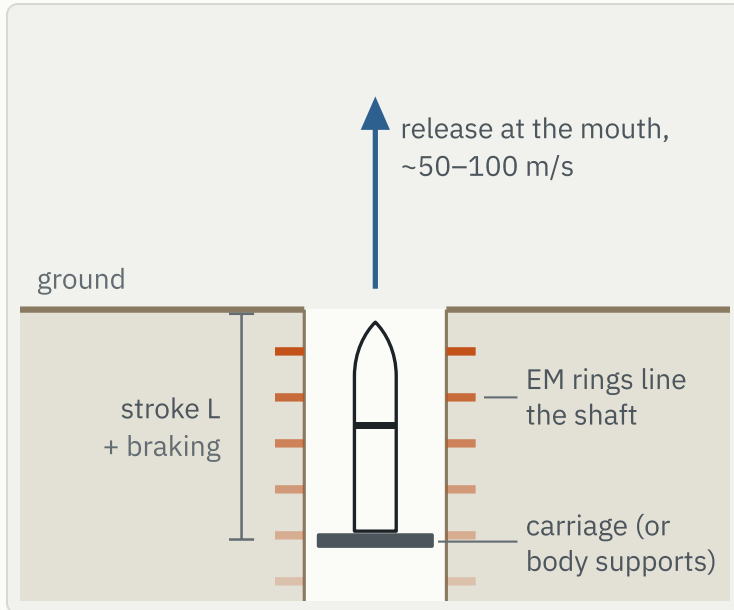
CAVEATS No structural mass for the 4 g push (~8.1 t of stage-1 strengthening cancels the gain, ~8.5 t the offload); gate vehicle +14.3% high; as pre-registered, the offload is mostly the lighter stack, not the push (slide 14).

THE CONCEPTS

Three ways to start: a silo, a ramp, and the pad baseline

MODELLED · CONCEPT A ONLY SO FAR

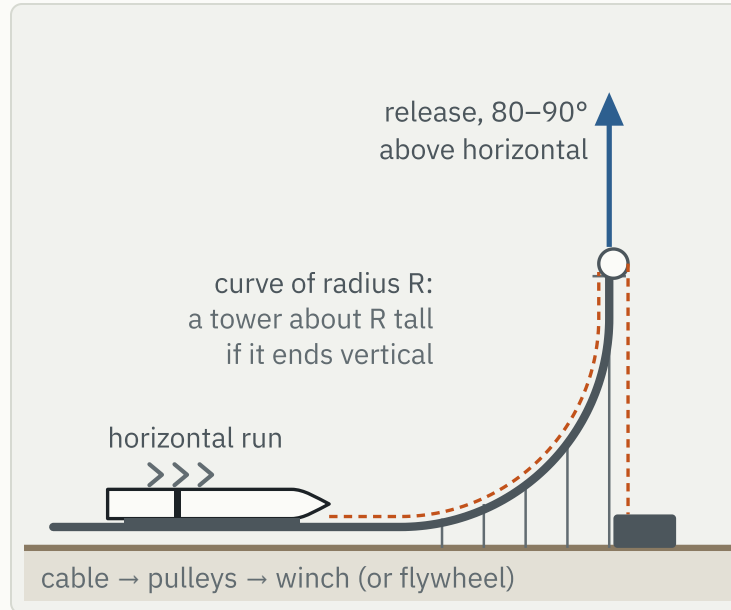
A · VERTICAL MAGLEV SILO



Rings in an underground shaft form a vertical linear motor. Engines light in the shaft (hot) or just after the exit (cold). Modelled as a prescribed-acceleration push.

PLANNED · README PHASE 3

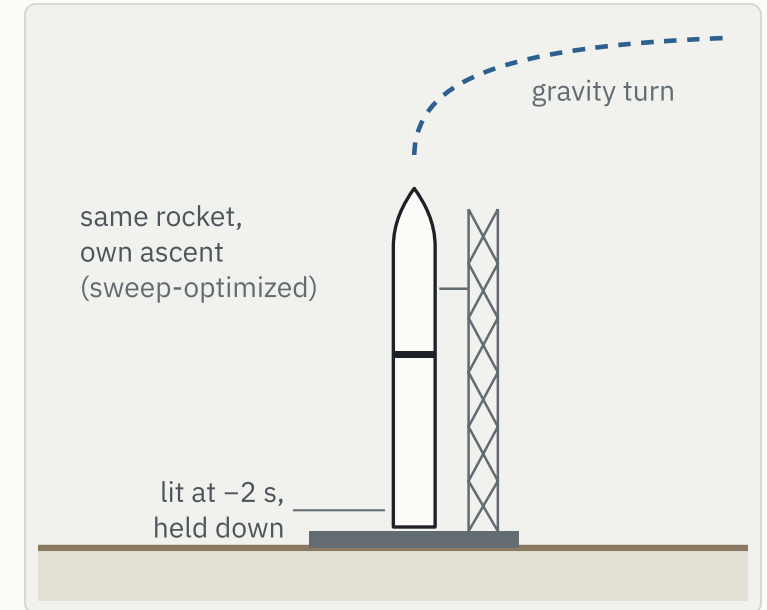
B · CABLE-PULLED RAMP



A winch pulls a carriage along a track that curves from horizontal toward vertical, and the rocket is released at the top. The original idea used a train to pull the cable.

BASELINE · EVERY RESULT IS A DELTA

C · PAD BASELINE



The same rocket from a normal pad, with its own sweep-optimized ascent in the same guidance family. Every result is a difference against C.

Every metre per second from the ground should help

Any ground-supplied velocity, however small, cuts the propellant the rocket needs, or raises its payload. The ground energy can come from renewables and be reused every launch.

README.md, "The concepts"

HOW IT IS TESTED

- Every run is a difference against the same rocket launched from a pad.
- Pad and variants share the vehicle, the guidance family, the sweep grid and the search budget.
- No physics feature is used in an experiment until its analytic test passes.
- Results that undercut the hypothesis are reported as plainly as the ones that support it.

WHAT COULD SINK IT

- Extra structure for a full, fuelled stack pushed at 4 g felt.
- Speed lost while the engines light after release.
- Aerodynamic loads from starting fast, low in the atmosphere.
- Ground power: gigawatt-class peaks for a few seconds.

"The simulator's real job is to measure these penalties against the ideal numbers" (README).

About 9 kg of payload per m/s, in the ideal case

HAND CALCULATIONS · MADE BEFORE THE SIMULATOR WAS BUILT

IDEAL ROCKET EQUATION · LOSSES HELD CONSTANT

EXIT SPEED $v = \sqrt{(2aL)}$

TRACK LENGTH	1 G	3 G	5 G
10 m	14 m/s	24 m/s	31 m/s
50 m	31 m/s	54 m/s	70 m/s
100 m	44 m/s	77 m/s	99 m/s
300 m	77 m/s	133 m/s	172 m/s
1,000 m	140 m/s	243 m/s	313 m/s

a is the net acceleration along the track; a vertical silo must also supply 1 g to hold the weight. **Simulation update:** the silo model reproduces $\sqrt{(2aL)}$ exactly, 76.71 m/s after 2.607 s at 3 g net over 100 m (RQ3-1d).

IDEAL SCREENING: WHAT AN ASSIST IS WORTH

ASSIST	PAYLOAD GAIN	OR STAGE-1 PROPELLANT SAVED
25 m/s	+220 kg (+1.0%)	—
50 m/s	+440 kg (+1.9%)	—
77 m/s	+690 kg (+3.0%)	14 t (3.6%)
99 m/s	+890 kg (+3.9%)	18 t (4.6%)
150 m/s	+1,360 kg (+6.0%)	—
300 m/s	+2,790 kg (+12.2%)	53 t (13.5%)

Generic Falcon 9-class two-stage vehicle, 542.6 t (published 549 t); stage-1 dry mass 25.6 t and average Isp 295 s assumed; fairing dropped at staging. NASA's "over 20%" fuel-saving claim for ~275 m/s is a benchmark to explain, not a target. **Simulation update:** at 76.7 m/s the 2-D gate vehicle (+14.3% high) leaves out 41.26 t, 2.76× its own 14.98 t screening offload, before structural mass; part of the beat is the lighter stack and the pad's hold-down burn (slides 13–14).

Nine penalties the simulator has to measure

HAND the README's first-order size · **SIM** measured since (preliminary, concept A, sweep-optimized, gate vehicle +14.3% high) · also in the README: train traction, sideways loads

AXIAL LOAD ON A FULL STACK

HAND A 3 g net push means a fuelled stack feels 4 g: 21.5 MN at the interface, 2.8× liftoff thrust.

SIM 3.996 g and 22.49 MN on the 2-D stack. No structural mass charged.

LIGHTING AFTER RELEASE

HAND $g \times (\text{delay} + \text{ramp}/2)$: 5–25 m/s, up to a third of a 77 m/s assist.

SIM 2-D costs about 2× more: 0.5 s + 2 s ramp loses 32.30 m/s on the README masses.

HARDWARE AND STRENGTHENING

HAND Break-even at ~5.5 t added to stage 1, ~0.7 t to stage 2.

SIM ~184 kg of payload per tonne of stage-1 mass, so break-even ~8.1 t; the 41.26 t offload loses 4.5–5.1 t per tonne, zero near 8.5 t.

PEAK POWER

HAND ≈2.2 GJ in 2.6 s, ~1.65 GW peak; ~1.2 MWh of electricity at 50%.

SIM 2.249 GJ with a 1.725 GW peak on the heavier 2-D stack.

AERODYNAMICS

HAND Extra speed low in dense air raises drag and possibly max-Q.

SIM Unthrottled max-Q –16.0% at full load, +3.4% at the offload; drag loss –3.5 m/s; q-alpha at the kick 1.7–3.4× the pad's.

FAILED IGNITION (A)

HAND An unlit rocket coasts ~300 m up, then falls back into the silo.

SIM Back at the mouth at 76.60 m/s after 15.689 s. No abort modelled.

SILO AIR (A)

HAND A snug shaft is a piston: up to 1.1 MN for a 3.7 m bore, ~5% of the drive.

SIM Every run vents the shaft, with no shaft drag (biases power ~0.08% low).

CARRIAGE BRAKING

HAND At 77 m/s, braking at 5 g needs ~60 m.

SIM 60.0 m added: the 100 m stroke needs a 160 m facility.

CURVES AND SHALLOW RELEASE (B)

HAND $R = v^2/a_n$: 600 m for 1 g at 77 m/s; an unwinged release droops 5.2°/s at 45°.

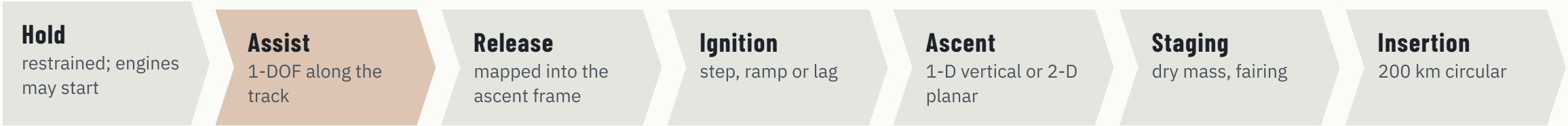
SIM Not modelled yet: concept B waits for README Phase 3.

Launch assist is not a new idea

PROJECT	WHAT IT DID	STATUS	LESSON
NASA Magnetic Launch Assist / MagLifter (1990s–2000s)	Horizontal maglev track about 1.5 miles long, 600 mph in 9.5 s	Technology demos only	The closest precedent; its claims are a benchmark to explain
EMALS, U.S. Navy	91 m linear motor launches a 45 t aircraft to 240 km/h; four 121 MJ flywheels	Operational	Proves ~100 m, ~2.5 g, ~70 m/s at 45 t
Holloman maglev sled, USAF (2016)	Rocket-propelled superconducting sled, 633 mph on 2,100 ft	Test facility	Maglev works for sleds near Mach 1
DARPA/NASA Horizontal Launch Study (2011)	Reviewed 130+ horizontal-launch studies spanning 60 years	Only Pegasus became operational	Many concepts close on paper; cost decides
NASA KSC railgun launch assist (2011)	Argued for 2–3 g for several seconds, beyond Mach 1	Analysis, lab tests	Another drive option
Radian Aerospace	~2-mile rocket sled releases a winged spaceplane at Mach 0.7, engines lit	In development; subscale tests 2024	Wings make a shallow release workable
SpinLaunch	33 m vacuum centrifuge, ~10,000 g, 10 test flights (2021–22)	No orbital system built (2025 reports)	Kinetic launch suits only g-hardened payloads
Long March 11, China	58 t solid launcher cold-launched from a tube, igniting after ejection	Operational	"Eject, then ignite" works at 58 t with solids
Intamin Accelerator Coasters (Kingda Ka)	Hydraulic winch and cable: 0–206 km/h in 3.5 s, then a 139 m vertical tower	Kingda Ka demolished 2025	Concept B at park scale, rollbacks included

This project's angle is narrow: measure, on one simulator, what a push is worth to an unchanged two-stage rocket, always against the same rocket from a pad. Pad-start tools (Kerbal Space Program, OpenRocket, RocketPy) cannot ask that question.

Phases joined by events, and every piece tested first



BUILT IN PHASES 0-2

- 2-D planar ascent over a spherical rotating Earth; a 1-D vertical model.
- ICAO standard atmosphere, drag varying with Mach, back-pressure.
- Vertical rise, kick, gravity turn; linear-tangent steering on stage 2.
- Payload capacity found by bisection; guidance sweep-optimized.
- Constant-acceleration silo push, with no force or power limit.

VALIDATION FIRST

- Rocket equation in vacuum: relative error < 1e-6.
- Elliptical orbit ($e \approx 0.3$), 10 revolutions: energy and momentum drift < 1e-8.
- Equatorial east pad starts at 465.1 m/s inertial.
- Loss budget closes to < 0.01 m/s over a full ascent.
- Assist energy identity < 1e-6; 10× tighter tolerances move payload < 0.1%.

TESTS AND OUTPUTS

200	269	951
tests at the Phase 0 gate	at the Phase 1 gate	928 fast + 23 slow at the Phase 2 handoff

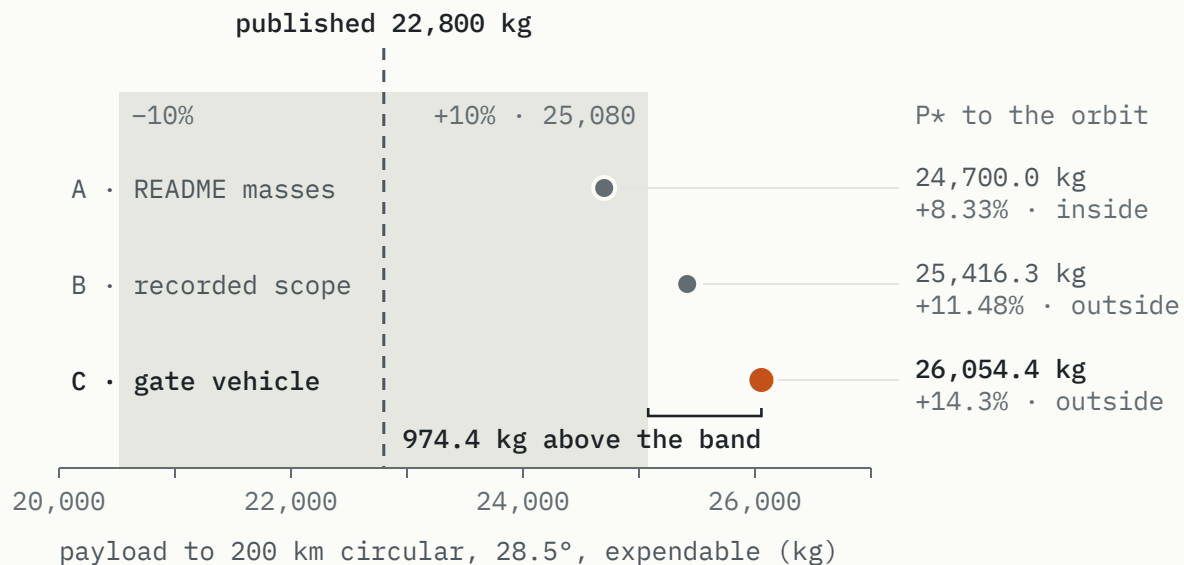
Every run writes `summary.md`, `metrics.json`, CSV time series and plots. `animate` makes an MP4 or GIF; `replay` an interactive HTML page.

CALIBRATION

The gate vehicle lands +14.3% high: a documented miss

LABELLED CALIBRATION, NOT VALIDATION

200 KM CIRCULAR · 28.5° · DUE EAST · EXPENDABLE



The reference orbit was fixed before the run and never moved. Nothing was tuned toward the published 22,800 kg.

The pre-registered gate vehicle (mass set C) reaches **26,054.4 kg**, 974.4 kg above the $\pm 10\%$ band's upper edge of 25,080 kg.

Every numerical check passes. The cause is not established, because the omissions point both ways: throttling, reserves and residuals would lower the payload; better guidance could raise it.

The user accepted the miss as documented on 2026-09-30. **Every 2-D number on the next slides is on the gate vehicle and inherits it.**

WHAT IT MEANS The miss carries into every 2-D gain. On a vehicle calibrated to 22.8 t, the cold-start silo's gain would be expected near 1.3 t instead of 1.5 t (derived by proportion, not run).

To orbit, the cold-start silo gains +1,498.8 kg (+5.75%)

PRELIMINARY FINDING · RQ3-2D, RQ2-2D

CONCEPT A · 3 G NET OVER 100 M · RELEASED AT 76.7 M/S

RUN	STAGE-1 START	PAYLOAD P*	GAIN OVER PAD	
pad (baseline)	lit at −2 s, 2 s ramp, held down	26,054.4 kg	–	
pad_instant (unphysical yardstick)	full thrust at release	26,137.5 kg	+83.1 kg	
silo_instant (unphysical yardstick)	full thrust at release	27,880.4 kg	+1,826.0 kg	
silo_cold (reference)	lit 0.5 s after release, 2 s ramp	27,553.2 kg	+1,498.8 kg	
silo_hot_ramp_on_track	lit on the carriage, full thrust at release	27,788.2 kg	+1,733.8 kg	
silo_hot_full	full thrust for the whole push	27,543.0 kg	+1,488.6 kg	

Hatched bars: unphysical yardsticks (a step to full thrust). Bars scale to the largest gain, 1,826.0 kg.

+1,498.8 kg

+5.75% over the pad's 26,054.4 kg

Under every ±10% sensitivity case the gain stays between **+1,345** and **+1,667 kg**.

Best physical variant: lit on the carriage, ramp ending at release, **+235.0 kg** over the cold start, paid for with 2.7 t burned on the track (no sensitivity run).

CAVEATS Sweep-optimized guidance, not optimal control · unthrottled, so max-Q and q-alpha are upper bounds · free kick (no angle-of-attack aerodynamics) · gate vehicle +14.3% high · no structural mass for the 4 g push · prescribed-acceleration drive, massless carriage, no shaft drag · planar point mass, generic C_D(Mach).

Twice the ideal estimate, and the loss breakdown says why

GAIN AGAINST THE IDEAL SCREENING

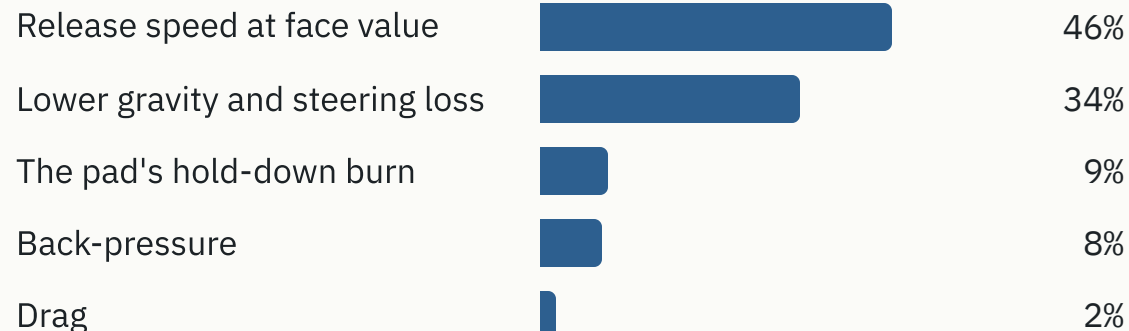


1.96×, about 2.0×. The project's rule: an assist that beats the screening must be explained by the loss breakdown, or it is treated as a bug.

A vertical release is not constant-loss. In silco_cold the gravity, steering, back-pressure, drag and fairing terms together fall by **74.1 m/s**, and the pad pays **14.4 m/s** for burning 2.7 t while clamped. An early vertical m/s is worth about 22.7 kg of payload on this vehicle (release anchor), 2.3× the screening's 10 kg.

BUT SMALL PUSHES gain little from the push itself. At the slow corner of the sweep (0.5 g, 50 m, 22 m/s) the trajectory term is negative (−11.1 m/s), and about 127 kg gross (about 83 kg net, 35%) of that point's +240 kg is the pad's hold-down burn, not the push.

WHERE THE GAIN COMES FROM (MATCHED BASIS)

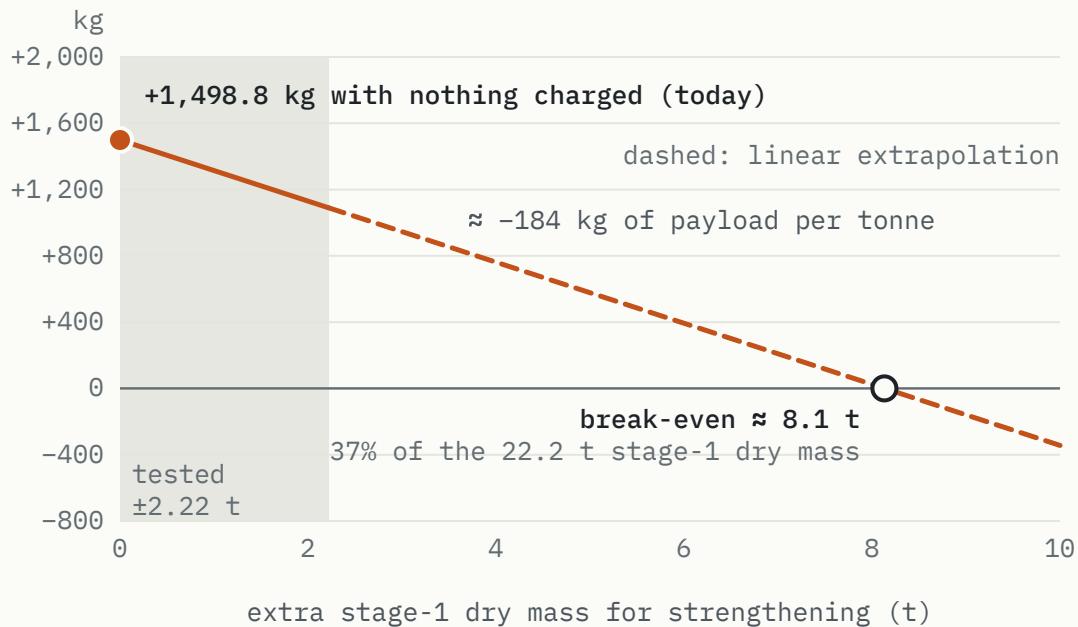


Capacity basis: 51 / 27 / 10 / 9 / 2%. Nothing is left unexplained beyond search noise.

READ WITH The 34% trajectory term appears only after MECO. It is the part a better (Phase 5) pad ascent could change, and optimal control could move it in either direction. The 9% is a baseline convention, not the push.

What goes against it, stated as plainly

STRUCTURAL MASS COULD CANCEL THE WHOLE GAIN



Derived from the $\pm 10\%$ stage-1 dry-mass cases of silo_cold; a linear extrapolation beyond them, not a sized structure. No mass is charged today for the 4.0 g full-stack load (22.5 MN at the interface).

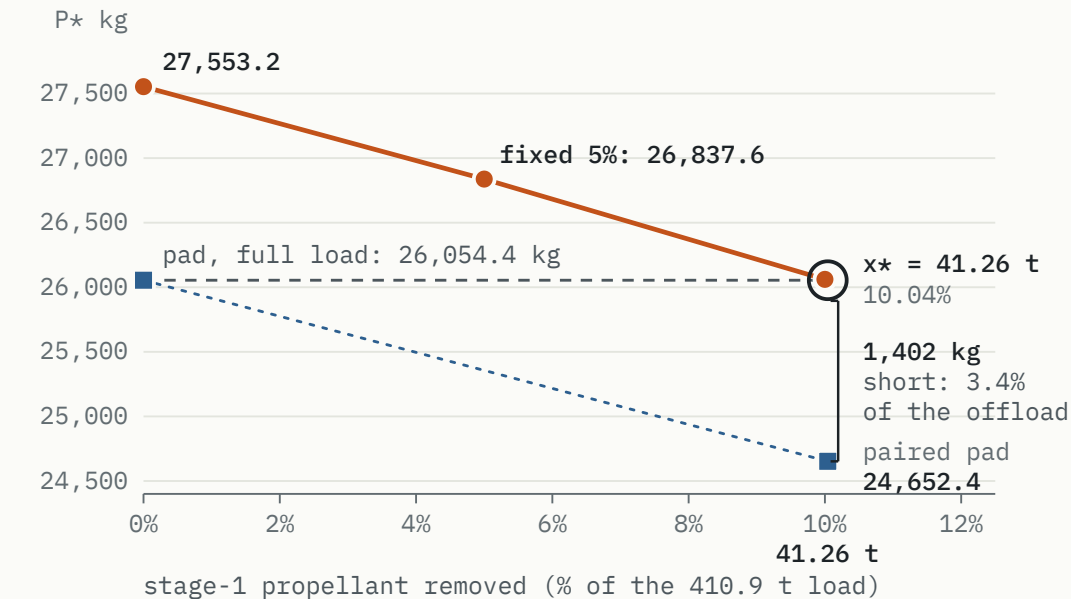
1. **The startup after release is expensive.** The cold start gives back 327.1 kg (34.74 m/s) against an instant start, $2.37\times$ the 1-D formula. A 3 s lag after the same 0.5 s delay gives back 781 kg.
2. **A full hot start does not pay.** silo_hot_full ends 10.2 kg behind the cold start, and -18.5 to $+0.1$ kg under the $\pm 10\%$ cases: a tie, not a gain.
3. **Part of the headline is the baseline.** The pad's 2.7 t clamped ramp accounts for 130.7 kg gross (83.1 kg net). Against a pad lit at release the gain is +1,415.8 kg.
4. **The calibration miss inflates it.** On a vehicle calibrated to 22.8 t the gain would be near 1.3 t (derived by proportion, not run).
5. **One aerodynamic load gets worse.** q-alpha at the kick is $1.7\text{--}3.4\times$ the pad's (130–251 Pa rad at 77 m/s), with no penalty charged.
6. **There is no abort.** After a failed ignition the full stack falls back to the shaft mouth at 76.6 m/s after 15.7 s.
7. **Small pushes gain little from the push itself.** At 0.5 g over 50 m (22 m/s) the trajectory term is negative (-11.1 m/s), and about 83 kg net (35%) of the +240 kg is the pad's hold-down burn.
8. **The trajectory gain is unverified** beyond the gravity-turn family: it appears only after MECO, and optimal control could move it either way (slide 11).
9. **The facility is sized by power.** A 1.725 GW peak at release after a 2.6 s push, 2.249 GJ per launch, and 160 m of facility with the braking.

41.26 t less stage-1 propellant, before structural mass

PRELIMINARY FINDING · RQ1-FUEL-OFFLOAD-2D

PRE-REGISTERED · RUN 2026-10-03

CONCEPT A · 3 G NET OVER 100 M · COLD START



—● silos cold start ○ solved x^* ■ pad (full; same offload) ---- P_{ref}

Lines join the recorded runs. The fixed 5% and 10% cases fly their own payload capacity (+783.2 and +6.8 kg over P_{ref}); x^* flies exactly P_{ref} .

41.26 t

of stage-1 propellant: **10.04%** of the 410.9 t stage-1 load, **7.96%** of the 518.4 t total

- Two different vehicles, the same payload (the pad's 26,054.4 kg) to the same 200 km orbit at 28.5°. Released at 76.7 m/s, lit 0.5 s later, 2 s ramp.
- Checked: the offloaded vehicle's own payload search returns the pad's payload (+0.0085 kg); the pad's own stage-1 control removes nothing.
- $\pm 10\%$ sensitivity arms: **38.67–44.46 t**. 2.76× the ideal-screening estimate (14.98 t); the loss breakdown explains why (next slide).

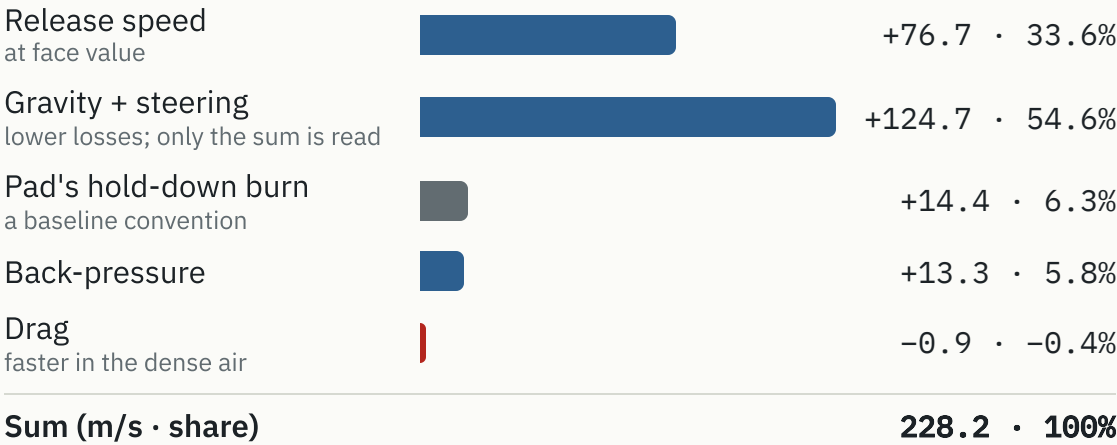
THE BRIDGE · README MASSES, +8.3%

Calibrated inside the band: **36.01 t**, **9.10%** of its 395.7 t stage-1 load. Read it as 9–10% of stage 1 across the two models, not as a Falcon 9 figure.

CAVEATS Gate vehicle, +14.3% high · as pre-registered, most of the offload is the lighter stack's thrust-to-weight (the paired pad is only 1.4 t short); a delta-v reading chosen after the run gives the lighter stack 28–29% (slide 14) · sweep-optimized, unthrottled, free kick · no structural mass for the 4 g push (slide 15) · max-Q 3.4% above the pad's · prescribed 3 g drive, massless carriage, vented shaft · partly filled tanks, dry masses unchanged · stage 1 is the only headline (stage 2 failed its verification).

Where the 41.26 t comes from, and how much of it is the push

IDEAL DELTA-V THE OFFLOADED VEHICLE DOES WITHOUT



Both vehicles fly the pad's payload; the split closes to 4.5e-12 m/s, so the beat over the screening is explained, not a bug. The offloaded vehicle reaches MECO 12.8 s sooner and loses 1,367.2 m/s to gravity against the pad's 1,492.0.

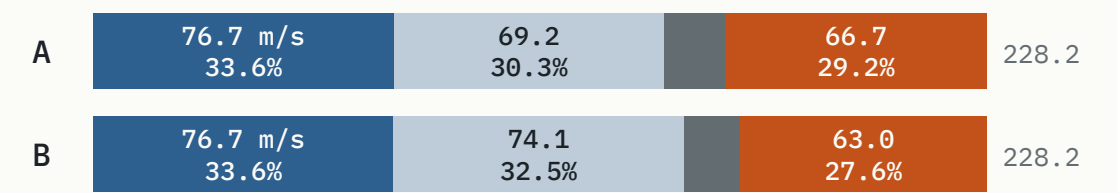
READ WITH The trajectory terms are unverified beyond the gravity-turn family: optimal control (README Phase 5) could move them either way. Fairing -0.02 m/s, margin 0.

HEAD START OR LIGHTER STACK? THE PAIRED PAD

The pad flown with the same 41.26 t offload and no push carries **24,652.4 kg, 1,402.0 kg short**. The push is needed to carry the offload.

AS PRE-REGISTERED That shortfall is 3.4% of the offload, which is small, so the branch fires: **most of the offload is the lighter stack's thrust-to-weight**, not the head start. The note reports this verdict, and argues that it compares payload kilograms with propellant kilograms.

READ AFTER THE RUN In ideal delta-v, a reading the pre-registration does not spell out, the 228.2 m/s splits as follows (two orders of the split):

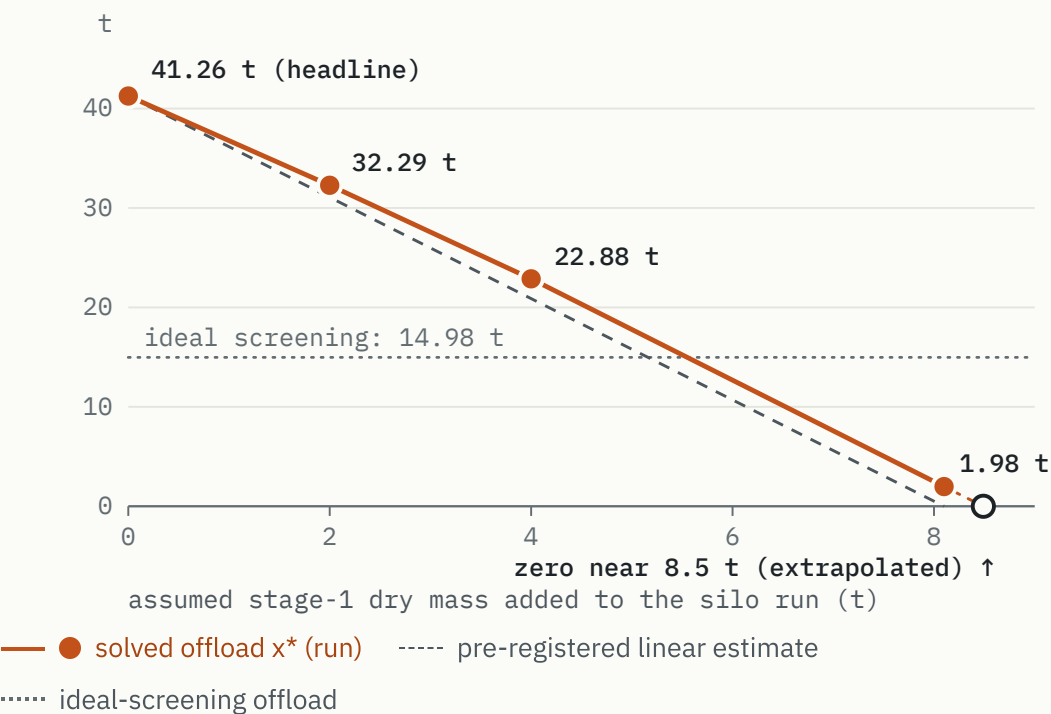


■ release speed ■ head start's trajectory effects ■ hold-down (6-7%) ■ lighter stack

The lighter stack's own thrust-to-weight gain is 28-29%; the push and its trajectory effects 64-66%. Order A goes through the paired pad, B through the full-load silo. Derived by difference between runs.

About 8.5 t of assumed stage-1 structure cancels it

OFFLOAD LEFT AFTER AN ASSUMED STAGE-1 DRY MASS



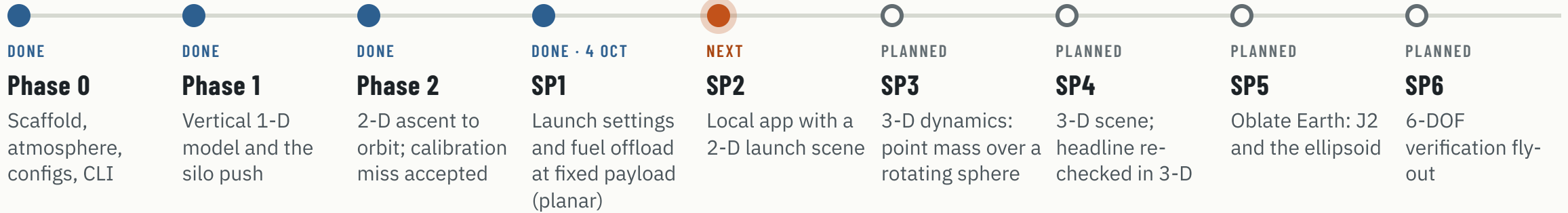
ASSUMED STAGE-1 DRY MASS	OFFLOAD LEFT	% OF STAGE 1	MAX-Q VS PAD
none (headline)	41.26 t	10.04%	+3.35%
+2 t	32.29 t	7.86%	-2.18%
+4 t	22.88 t	5.57%	-7.57%
+8.1 t	1.98 t	0.48%	-18.19%
≈ 8.5 t (extrapolated)	0	0	—

- Each tonne takes 4.5–5.1 t off, and the erosion steepens; every row lies above the pre-registered linear estimate. 8.5 t is 38% of the 22.2 t stage-1 dry mass.
- **Loads at the headline.** Max-Q 38.4 kPa, 3.35% above the pad's: the full-load silo's 16.0% reduction is spent. q-alpha at the kick is 3.0× the pad's, and the stack feels 4.0 g on the track (20.81 MN). Unthrottled.
 - **Energy, not an efficiency claim.** The 12.4 t of RP-1 removed carry about 128× the push's 1.16 MWh of electricity in combustion heat (5.65× at +8.1 t); the ratio leaves out LOX production, fuel supply, grid losses and the facility.
 - **Stage 2 is not a headline.** Its case failed independent verification (31.90 t gross, a flagged lower bound).

ASSUMPTIONS The rows add an assumed dry mass to the silo run only; they are not a sized structure. No structural model exists yet: README Phase 3 decides whether the headline survives.

ROADMAP

One phase per session, each ending in something you can run



README PHASE 3 · ASSIST MODELS

Force- and power-limited linear motor, curved ramp, cable winch, air column, modelled braking, and the structural mass for the 4 g push. **This decides whether the headline survives:** about 8.5 t of assumed stage-1 dry mass cancels SP1's 41.26 t (extrapolated). Not scheduled; can be moved up at any session boundary.

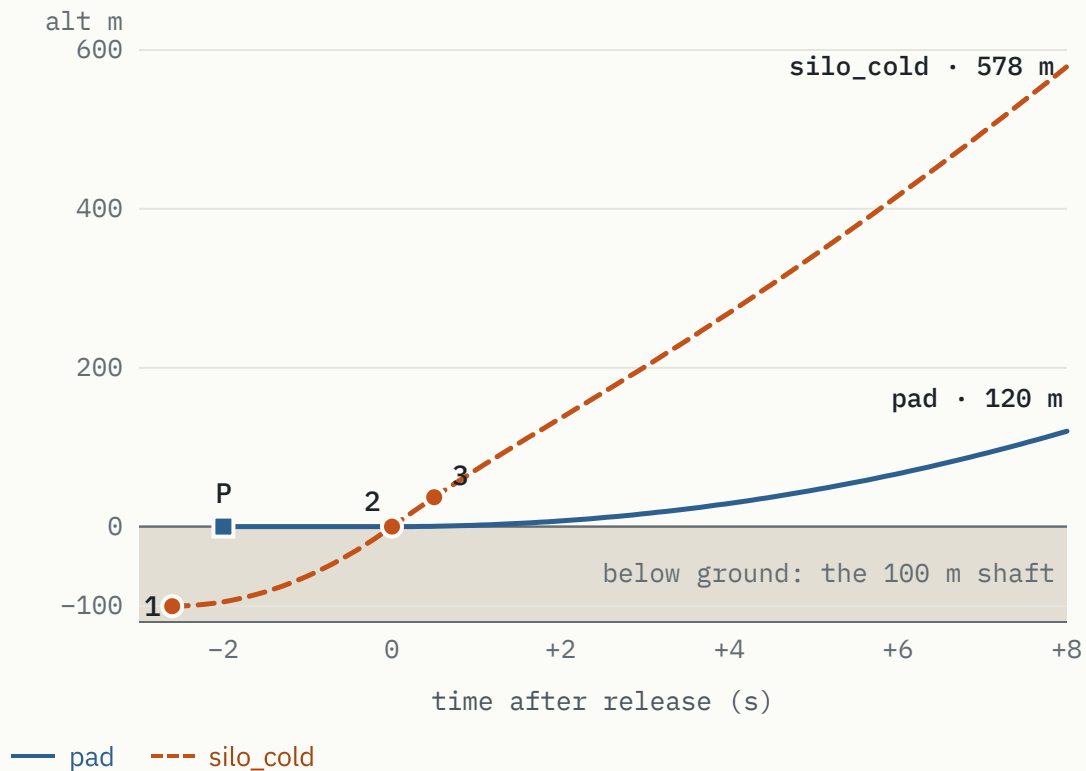
README PHASE 5 · FAIR COMPARISON

Optimal-control ascent for every configuration; the headline results re-run with optimized guidance instead of sweep-optimized.

README PHASE 6 · HOBBY SCALE

A RocketPy 6-DOF model of a real rocket with an assist before the rail; baseline apogee within $\pm 5\%$ of RocketPy.

The head start, in the first 8 seconds of the recorded run



1 The push starts 100 m down: 3 g net, 4.0 g felt.

2 Release at the silo mouth at 76.7 m/s.

3 Lit 0.5 s later, 37 m up, with a 2 s ramp.

P The pad: lit at -2 s and held down until release.

At 8 s after release the silo stack is at 578 m and 85 m/s; the pad at 120 m and 31 m/s.

ANIMATION EXAMPLES

- **SP1 fuel offload:** the pad against the offloaded silo at the same payload and orbit, and the offloaded silo against its paired pad (interactive replays).
- **Pad against cold-start silo:** both flown to a 200 km orbit.
- **Ignition timing:** four ways to light stage 1.
- **Failed ignition:** the push with no engines, back at the mouth at 15.7 s.

Videos, GIFs and interactive replays: khoks.github.io/.../examples/

Point-mass planar model, sweep-optimized and unthrottled, gate vehicle +14.3% high, no structural mass for the 4 g push. A model replay, not a design result.

READ MORE

Where the full numbers and their caveats live

ON GITHUB

[github.com/khoks/
RocketLaunchOptimizationSimulator](https://github.com/khoks/RocketLaunchOptimizationSimulator)

README.md: the research brief, prior art, hand numbers and results so far.

docs/findings/: one note per research question, caveats first.

docs/phases/: the program board, SP1 to SP6.

ON THE SITE

[khoks.github.io/
RocketLaunchOptimizationSimulator](https://khoks.github.io/RocketLaunchOptimizationSimulator)

User manual: install, run, experiment and vehicle files, outputs.

Animation examples: replays of the recorded 2-D runs.

This deck, also as a [PDF](#).

HOW IT IS RUN

From the user manual (Python 3.12 and uv):

```
uv sync
```

```
uv run python -m launchsim run  
experiments/silo_screening_2d.yaml
```

```
uv run python -m launchsim replay  
results/silo_screening_2d/<timestamp>  
--runs pad silo_cold
```

Reading is free. Any other use needs written permission (next slide).

FINDINGS: THE SUGGESTED READING ORDER

CAL-f9-leo-2d

The vehicle every 2-D number rests on, and its +14.3% miss.

RQ1-fuel-offload-2d

Propellant replaced at fixed payload: the headline, its decomposition and penalty rows.

RQ3-silo-screening-2d

The payload gain, the screening check and the full caveat list.

RQ2-ignition-timing-2d

Hot or cold start; what delay, ramp and lag cost.

RQ6-aero-2d-preliminary

Max-Q, q-alpha and drag, unthrottled.

RQ3-1d, RQ2-1d

The 1-D record: closed forms, loads, energy, failed ignition.

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github.com/khoks/RocketLaunchOptimizationSimulator